

College of Industrial Technology  
King Mongkut's University of Technology North Bangkok

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| Seat No. |

Midterm Examination of Semester 1

Year: 2014

Subject: 392153 Chemistry III

Section: 15 - 16

Date: 8 October 2014

Time: 13.00 - 15.00

Name: \_\_\_\_\_ ID: \_\_\_\_\_ Class: \_\_\_\_\_

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Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
  2. No documents are allowed to be taken out of the examination room.
  3. Textbooks, dictionaries and calculator are **NOT** allowed.
  4. Electronic communication devices are **NOT** allowed in the examination room.
  5. The examination has 8 pages (including this page and periodic table), total score of 80 points.
  6. Write all your answers on this examination sheet.
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1. Please give shortly correct explanation of statements. (10 points)

Statements

- (I) The molecule  $\text{SO}_2$  has a net dipole.

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- (II)  $\text{NO}_3^-$  and  $\text{CO}_2$  are planar species.

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- (III)  $\text{MgO}$  has high melting point.

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- (IV)  $\text{BF}_3$  has a tetrahedral geometry.

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- (V) Hydrogen bond is stronger than covalent bond.

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## 2. Please give full explanation of statements (30 points)

| (I) | Element | Electronegativity |
|-----|---------|-------------------|
|     | H       | 2.1               |
|     | Cl      | 2.8               |
|     | Mg      | 1.2               |
|     | N       | 3.1               |
|     | F       | 4.1               |

Given the electronegativities above, which of the following compounds has the bonds with the greatest polarity? Why?

HF      NH<sub>3</sub>      HCl      MgCl<sub>2</sub>      F<sub>2</sub>

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(II) Of the following compounds, which has the highest boiling point?

CH<sub>3</sub>—CH<sub>3</sub>      CH<sub>3</sub>—CH<sub>2</sub>OH      CH<sub>3</sub>—COOH      CH<sub>3</sub>—COH      H<sub>2</sub>S

(III) In  $\text{H}_2\text{O}$ ,  $\text{H}_2\text{S}$ ,  $\text{H}_2\text{Se}$ ,  $\text{H}_2\text{Te}$ , the bond angle decreases though all have the same bent shape. Why?

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3. Calculate the formal charge on each atom in the following Lewis structures.  
(20 points)

(I) Methoxide ion ( $\text{CH}_3\text{O}^-$ )

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(II) Oxonium ion ( $\text{CH}_3\text{OH}^+$ )

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(III) Sulfuric acid ( $\text{H}_2\text{SO}_4$ )

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(IV) Nitromethane ( $\text{CH}_3\text{NO}_2$ )

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4. Use the data provided below to calculate the lattice energy of RbCl. Is this value greater or less than the lattice energy of NaCl (–786 kJ) ? Explain? (20 points)

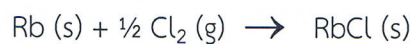
Electron affinity of Cl = –349 kJ/mol

1st ionization energy of Rb = 403 kJ/mol

Bond energy of  $\text{Cl}_2$  = 242 kJ/mol

Sublimation energy of Rb = 86.5 kJ/mol

$\Delta H_f [\text{RbCl (s)}] = -430.5 \text{ kJ/mol}$





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Honesty is the most important...



# Periodic Table of Elements

| 1                                           | 2                                       | 3                                        | 4                                          | 5                                        | 6                                       | 7                                          | 8                                      | 9                                       | 10                                        | 11                                       | 12                                       | 13                                      | 14                                     | 15                                        | 16                                       | 17                                        | 18                                       |
|---------------------------------------------|-----------------------------------------|------------------------------------------|--------------------------------------------|------------------------------------------|-----------------------------------------|--------------------------------------------|----------------------------------------|-----------------------------------------|-------------------------------------------|------------------------------------------|------------------------------------------|-----------------------------------------|----------------------------------------|-------------------------------------------|------------------------------------------|-------------------------------------------|------------------------------------------|
| 1<br><b>H</b><br>Hydrogen<br>1.00794        | 2<br><b>He</b><br>Helium<br>4.002602    |                                          |                                            |                                          |                                         |                                            |                                        |                                         |                                           |                                          |                                          |                                         |                                        |                                           |                                          |                                           |                                          |
| Atomic #<br>Symbol<br>Name<br>Atomic Weight |                                         |                                          |                                            |                                          |                                         |                                            |                                        |                                         |                                           |                                          |                                          |                                         |                                        |                                           |                                          |                                           |                                          |
| 3<br><b>Li</b><br>Lithium<br>6.941          | 4<br><b>Be</b><br>Beryllium<br>9.012182 |                                          |                                            |                                          |                                         |                                            |                                        |                                         |                                           |                                          |                                          |                                         |                                        |                                           |                                          |                                           |                                          |
| 11<br><b>Na</b><br>Sodium<br>22.98976928    | 12<br><b>Mg</b><br>Magnesium<br>24.3050 |                                          |                                            |                                          |                                         |                                            |                                        |                                         |                                           |                                          |                                          |                                         |                                        |                                           |                                          |                                           |                                          |
| 19<br><b>K</b><br>Potassium<br>39.0983      | 20<br><b>Ca</b><br>Calcium<br>40.078    | 21<br><b>Sc</b><br>Scandium<br>44.955912 | 22<br><b>Ti</b><br>Titanium<br>47.867      | 23<br><b>V</b><br>Vanadium<br>50.9415    | 24<br><b>Cr</b><br>Chromium<br>51.9961  | 25<br><b>Mn</b><br>Manganese<br>54.938045  | 26<br><b>Fe</b><br>Iron<br>55.845      | 27<br><b>Co</b><br>Cobalt<br>58.933195  | 28<br><b>Ni</b><br>Nickel<br>58.6934      | 29<br><b>Cu</b><br>Copper<br>63.546      | 30<br><b>Zn</b><br>Zinc<br>65.38         | 31<br><b>Ga</b><br>Gallium<br>69.723    | 32<br><b>Ge</b><br>Germanium<br>72.65  | 33<br><b>As</b><br>Arsenic<br>74.92160    | 34<br><b>Se</b><br>Selenium<br>78.96     | 35<br><b>Br</b><br>Bromine<br>79.904      | 36<br><b>Kr</b><br>Krypton<br>83.798     |
| 37<br><b>Rb</b><br>Rubidium<br>85.4678      | 38<br><b>Sr</b><br>Strontium<br>87.62   | 39<br><b>Y</b><br>Yttrium<br>88.90585    | 40<br><b>Zr</b><br>Zirconium<br>91.224     | 41<br><b>Nb</b><br>Niobium<br>92.90638   | 42<br><b>Mo</b><br>Molybdenum<br>95.96  | 43<br><b>Tc</b><br>Technetium<br>(97.9072) | 44<br><b>Ru</b><br>Ruthenium<br>101.07 | 45<br><b>Rh</b><br>Rhodium<br>102.90550 | 46<br><b>Pd</b><br>Palladium<br>106.42    | 47<br><b>Ag</b><br>Silver<br>107.8682    | 48<br><b>Cd</b><br>Cadmium<br>112.411    | 49<br><b>In</b><br>Indium<br>114.818    | 50<br><b>Sn</b><br>Tin<br>118.710      | 51<br><b>Sb</b><br>Antimony<br>121.760    | 52<br><b>Te</b><br>Tellurium<br>127.60   | 53<br><b>I</b><br>Iodine<br>126.90447     | 54<br><b>Xe</b><br>Xenon<br>131.293      |
| 55<br><b>Cs</b><br>Caesium<br>132.9054519   | 56<br><b>Ba</b><br>Barium<br>137.327    | 57-71                                    | 72<br><b>Hf</b><br>Hafnium<br>178.49       | 73<br><b>Ta</b><br>Tantalum<br>180.94788 | 74<br><b>W</b><br>Tungsten<br>183.84    | 75<br><b>Re</b><br>Rhenium<br>186.207      | 76<br><b>Os</b><br>Osmium<br>190.23    | 77<br><b>Ir</b><br>Iridium<br>192.217   | 78<br><b>Pt</b><br>Platinum<br>195.084    | 79<br><b>Au</b><br>Gold<br>196.966569    | 80<br><b>Hg</b><br>Mercury<br>200.59     | 81<br><b>Tl</b><br>Thallium<br>204.3833 | 82<br><b>Pb</b><br>Lead<br>207.2       | 83<br><b>Bi</b><br>Bismuth<br>208.98040   | 84<br><b>Po</b><br>Polonium<br>(209)     | 85<br><b>At</b><br>Astatine<br>(209.9871) | 86<br><b>Rn</b><br>Radon<br>(222.0176)   |
| 87<br><b>Fr</b><br>Francium<br>(223)        | 88<br><b>Ra</b><br>Radium<br>(226)      | 89-103                                   | 104<br><b>Rf</b><br>Rutherfordium<br>(261) | 105<br><b>Db</b><br>Dubnium<br>(262)     | 106<br><b>Sg</b><br>Seaborgium<br>(266) | 107<br><b>Bh</b><br>Bohrium<br>(264)       | 108<br><b>Hs</b><br>Hassium<br>(277)   | 109<br><b>Mt</b><br>Meitnerium<br>(268) | 110<br><b>Ds</b><br>Darmstadtium<br>(271) | 111<br><b>Rg</b><br>Roentgenium<br>(272) | 112<br><b>Cn</b><br>Copernicium<br>(285) | 113<br><b>Uut</b><br>Ununtrium<br>(284) | 114<br><b>Fl</b><br>Flerovium<br>(289) | 115<br><b>Uup</b><br>Ununpentium<br>(288) | 116<br><b>Lv</b><br>Livermorium<br>(292) | 117<br><b>Uus</b><br>Ununseptium<br>(294) | 118<br><b>Uuo</b><br>Ununoctium<br>(294) |

C Solid

Hg Liquid

H Gas

Rf Unknown

Alkali metals

Alkaline earth metals

Lanthanoids

Actinoids

Transition metals

Poor metals

Other nonmetals

Noble gases

Metals

Nonmetals

57  
**La**  
Lanthanum  
138.90547 58 **Ce** Cerium 140.116 | 59 **Pr** Praseodymium 140.90765 | 60 **Nd** Neodymium 144.242 | 61 **Pm** Promethium (145) | 62 **Sm** Samarium 150.36 | 63 **Eu** Europium 151.964 | 64 **Gd** Gadolinium 157.25 | 65 **Tb** Terbium 158.92535 | 66 **Dy** Dysprosium 162.500 | 67 **Ho** Holmium 164.93032 | 68 **Er** Erbium 167.259 | 69 **Tm** Thulium 168.93421 | 70 **Yb** Ytterbium 173.054 | 71 **Lu** Lutetium 174.9668 | 103 **Lr** Lawrencium (262) || 89 **Ac** Actinium (227) | 90 **Th** Thorium 232.03806 | 91 **Pa** Protactinium 231.03588 | 92 **U** Uranium 238.02891 | 93 **Np** Neptunium (237) | 94 **Pu** Plutonium (244) | 95 **Am** Americium (243) | 96 **Cm** Curium (247) | 97 **Bk** Berkelium (247) | 98 **Cf** Californium (251) | 99 **Es** Einsteinium (252) | 100 **Fm** Fermium (257) | 101 **Md** Mendelevium (258) | 102 **No** Nobelium (259) | 103 **Lr** Lawrencium (262) |

For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.

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